Customer Segmentation Using K-Means and Gaussian Mixture Models

Abstract:

In this study, I researched how unsupervised machine learning can identify customer segmentation from demographic, spending, purchasing, and online experiment data. The dataset had 2,240 customers and 29 features. All the features were cleaned by removing records with missing values and transformed to create additional customer features. After that, eight numerical features were standardized and used to train K-means and Gaussian Mixture Model (GMM) clustering. We used the elbow method, selected four clusters for K-means, while the GMM BIC curve indicated eight components. K-means achieved a silhouette score of 0.1648 compared with 0.0702 for GMM. This result shows weak separation between customer groups, although K-means performed better than GMM. Later, if we can improve feature selection, outlier treatment, testing, and alternative clustering, we might be able to improve the result.

Introduction:

Customer segmentation is a process of grouping customers based on their similarities. It can help us understand the business. It can also help use understand the customers and whom to support more in the future. The following study, in this study, focuses on the following research question, which is How can K-Means and GMM clustering be used to identify distinct customer segments based on age, income, spending behavior, purchase frequency, and online engagement to support targeted marketing strategies? Meaning how using models like K-Means and GMM we can seperate a companies customers based on different features. Previous research has used clustering techniques to solve similar customer segmentation problems. [1] performed customer segmentation using the UCI Online Retail dataset and compared several clustering algorithms, including K-Means, GMM, and DBSCAN, using the Silhouette Score.[2] used Recency, Frequency, and Monetary (RFM) features and compared four clustering techniques. Their study found K-Means to be the best-performing method, with a Silhouette Score of 0.432619. [3]also used RFM-based clustering and found that customer segmentation based only on spending was insufficient, suggesting that multiple aspects of customer behavior should be considered.

Methodology:

Dataset Description:

Features: The dataset has 29 columns, including demographic information, product spending, purchase channels, and marketing campaign responses. Seven selected features will be used for clustering.

Target:

No predefined target variable because customer segmentation uses unsupervised learning. The Response column will not be used as a clustering target.

Size:

2,240 customer records and 29 columns, with 2,240 unique customer IDs.

Limitations:

There are 24 missing income values, extreme values in birth year and income, no existing customer segment labels, and historical customer records from 2012–2014.

Data cleaning:

1/Load the dataset using pandas and check for missing values and data types.

2/Remove the missing values

3/Convert Dt_Customer into datetime format.

4/Select the seven numerical features needed for clustering and exclude irrelevant columns.

Feature engineering:

Age: Calculate customer age using 2026 - Year_Birth.

Total_children: Combine Kidhome and Teenhome.

Total_spend: Calculate total spending across six product categories.

Customer_since: Calculate the number of days since customer enrollment.

Feature Scaling: Apply StandardScaler to the seven selected clustering features.

PCA:

Reduce the scaled data to two dimensions for visualizing customer clusters.

Models:

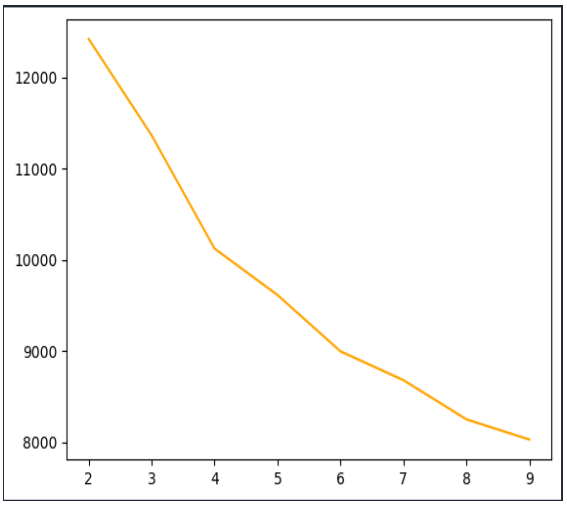
K-Means Clustering: The main model used to group customers based on similar demographic and purchasing behavior. The Elbow Method examines 2–9 clusters, and the notebook uses 6 clusters for the final model.

Gaussian Mixture Model (GMM) Groups customers using probability distributions and allows clusters to have different shapes and sizes.

Evaluation Metrics:

Elbow Method (Inertia): Helps determine a suitable number of clusters by measuring within-cluster variation. Lower inertia indicates more compact clusters but generally decreases as cluster count increases.

Result:



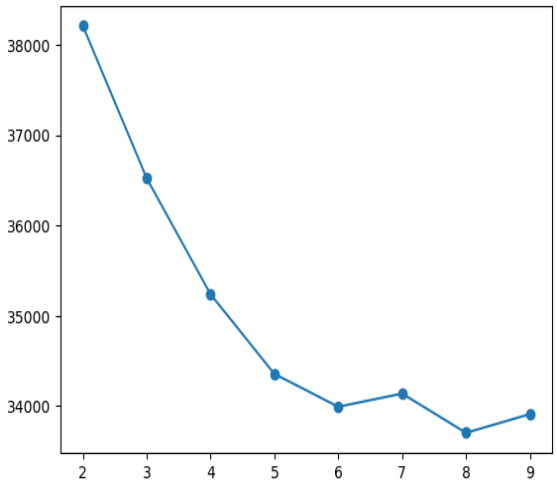
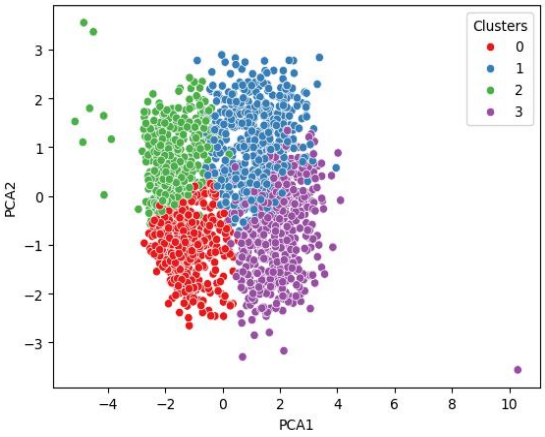


Fig1: K-Means Elbow method

Fig2: GMM Elbow method

Fig 1 shows the Elbow method for K-Means clustering Algorithm. From here we decided to take the value of 4 as cluster counts. Fig 2 shows the Elbow method for GMMclustering Algorithm. From here we decided to take the value of 8 as cluster counts.



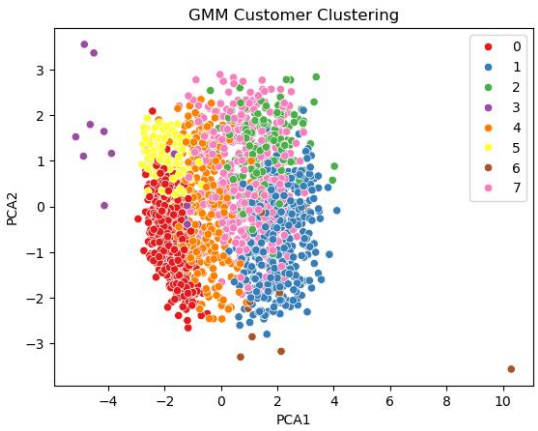
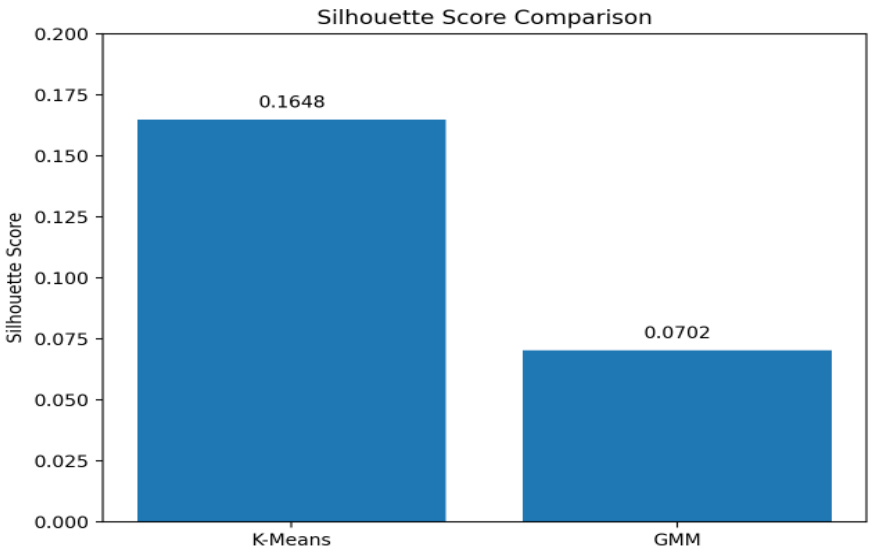


Fig2: K-Means Clusters

Fig4: GMM Clusters

Visual clusters after applying two different algorithms are shown here.



Final silhouette Score shows K-means give us better clusters in comparison to GMM.

Discussion:

The results indicate that **K-Means performed better than GMM** for the selected customer features. K-Means achieved a Silhouette Score of 0.1648, while GMM achieved 0.0702. This means that the K-Means clusters had better separation compared with the GMM clusters.

There are several limitations in this experiment. The dataset contains missing income values and extreme values in variables such as birth year and income. It also has no predefined customer segment labels, making external validation difficult. The records are historical, covering 2012–2014. The experiment removed missing values but did not perform a dedicated outlier-treatment step. In addition, the age and customer-tenure calculations use a 2026 reference date, which may not represent the customers' characteristics at the time of the original records.

Conclusion:

This study compared K-Means and GMM for customer segmentation using demographic, spending, purchasing, recency, online engagement, and customer-tenure features. K-Means selected four clusters using the Elbow Method and achieved a Silhouette Score of 0.1648, while GMM selected eight components using BIC and achieved 0.0702. K-Means therefore produced better-separated clusters than GMM for this dataset and feature set. However, both models showed weak cluster separation, indicating that further feature engineering, outlier treatment, and model experimentation would be useful for producing more clearly defined customer segments.

Reference:

[1] J. M. John, O. Shobayo, and B. Ogunleye, “An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market,” *Analytics*, vol. 2, no. 4, pp. 809–823, Dec. 2023, doi: 10.3390/analytics2040042.

[2] A. Ashraf, C. A. Rayed, N. A. Awad, and H. M. Sabry, “A Framework for Customer Segmentation to Improve Marketing Strategies Using Machine Learning,” *Procedia Comput. Sci.*, vol. 260, pp. 616–625, Jan. 2025, doi: 10.1016/j.procs.2025.03.240.
[3] O. Doğan, E. Aycin, and Z. Bulut, “CUSTOMER SEGMENTATION BY USING RFM MODEL AND CLUSTERING METHODS: A CASE STUDY IN RETAIL INDUSTRY,” *Int. J. Contemp. Econ. Adm. Sci.*, vol. 8, 2018, Accessed: Oct. 16, 2026. [Online]. Available: https://avesis.kocaeli.edu.tr/yayin/894de1af-d068-4e33-ad18-d0d727c24fbe/customer-segmentation-by-using-rfm-model-and-clustering-methods-a-case-study-in-retail-industry